When Does Length-Aware Attention Generalize in a Reduced Binary Classifier?

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Abstract

Softmax-attention classifiers trained on short sequences can fail to generalize to much longer ones, even on the simple task of detecting whether a target token is present, because a single target must compete for a fixed budget of attention against a crowd of non-target tokens that grows with the sequence. This report asks when making attention length-aware overcomes that dilution, in a reduced binary classifier where the mechanism is exactly analyzable: a single readout query attends over one-hot token embeddings, and because the non-target tokens are identical, the target's attention mass has an exact closed form. We compare constant, logarithmic, and learned length scalings, and find one condition behind all three: to generalize to unbounded length, the scaling must amplify the target's score margin fast enough to grow faster than the crowd of non-target tokens. A constant scaling never does; a logarithmic scaling does when the learned score margin is large enough, and a learned scaling does only when training makes the amplification fast enough. The consequence is diagnostic: a model can pass a large finite benchmark, such as ten million tokens, yet still be predicted to fail at greater lengths, because the quantity that decides unbounded-length generalization cannot be read off from any single benchmark.

1 Introduction

Length generalization is a basic difficulty for sequence models. A classifier can fit short training sequences while relying on a mechanism that does not remain valid at longer sequence lengths. This problem appears even in simple existential target-token tasks, where the model only needs to decide whether a sequence contains a particular token.

The motivating failure mode is attention dilution. Softmax attention divides a fixed budget of attention among all tokens, so a single target token must compete with a crowd of non-targets that grows with the sequence; even a fixed per-token margin for the target can be overwhelmed once that crowd is large enough. This raises a simple question: can attention be made length-aware, sharpening as the sequence grows, so that the target keeps enough attention at long lengths?

This report focuses on a reduced binary classifier for that question. The model is intentionally much simpler than a full transformer. It uses fixed one-hot token values, learned query and key projections, a single readout query (the query at the last position), and a binary classifier. The goal is not to show that this model is a competitive architecture. The goal is to test whether the simplified closed-form attention theory correctly describes a trainable model in the controlled setting where its assumptions can be directly checked.

The reduced model is built so that this theory applies exactly rather than approximately. Because every non-target token is identical, the attention scores collapse by construction to just two values, a target score and a shared non-target score. This two-score structure is precisely what the theory assumes, and it makes a closed-form expression for the target's attention mass exact, reducing length generalization to a precise question about how the target's score margin must grow with length.

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The report compares three scaling rules that govern how fast attention sharpens as the sequence grows: constant, logarithmic, and learned. The experiments play out as the theory predicts. Constant scaling improves with more training but always fails eventually. Log scaling succeeds once the score margin the model learns exceeds 1. Learned-log scaling succeeds at unbounded length only when optimization drives the product of the learned coefficient and the score margin above 1. This last regime carries the report's main lesson: a model can pass a finite benchmark, such as sequences of ten million tokens, yet still be predicted to fail at larger lengths whenever that product, which the benchmark alone cannot reveal, stays below 1. Passing a fixed length is therefore not evidence of unbounded-length generalization; the target's effective margin must grow faster than the competing crowd.

2 Related Work

This report describes the output of an independent research project in the Deep Network Understanding (DNU) Lab of Dickinson College. Consistent with the lab's emphasis on original research, this report relates its contribution to a small number of representative works rather than surveying the field comprehensively. We discuss two such works below, chosen for their direct connection to the mechanism studied here, and leave a fuller literature review to future work.

The transformer architecture of Vaswani et al. [2017] places softmax attention at its center. Each query forms a normalized distribution over the keys, so a fixed total attention mass is shared among all of them, and the attention given to any one token depends on how many keys it competes with. When the number of non-target keys grows with sequence length, this normalization is exactly the dilution effect studied here. The present report isolates that effect in a minimal setting where the attention scores and output values can be read off directly.

Press et al. [2022] study length extrapolation directly and propose attention with linear biases (ALiBi), which adds a fixed, distance-dependent penalty to the attention scores so that a model trained on short sequences can be evaluated on much longer ones. Their method operates in a full transformer and supplies a recency bias in place of standard positional encodings. The present report is narrower and more analytical. Rather than proposing a transformer method, it studies a length-aware multiplier on the scores, an inverse temperature that may be held fixed or learned, in a reduced classifier where the condition for length generalization can be derived in closed form and the learned score geometry can be inspected directly.

3 Background: Simplified Binary Attention

This section derives, from the model definition, the score structure and closed-form target-attention mass previewed in the Introduction, and then uses that closed form to characterize when the classifier generalizes to unbounded length.

The task uses a two-token vocabulary:

t target token;

u non-target token.

A positive length-n sequence, with $n \geq 2$, contains one target token and $n-1$ non-target tokens. The token in the last position is always the non-target token u. Because both positive and negative sequences end in u, this position carries no label information; it simply fixes a consistent readout slot. The model has no positional encoding, so permuting the tokens among the first $n-1$ positions does not change its output. We place the target in the first position only to make the notation easier to read; any position other than the last is equivalent:

```
t, u, u, ..., u
```

A negative length-n sequence contains only non-target tokens:

```
u, u, u, ..., u
```

At a high level, the query at the last position (the readout query) scores every token in the sequence, softmax converts those scores into attention weights, and the weighted sum of the token values is passed to a binary classifier.

The model begins with fixed one-hot embeddings for t and u:

$$t \mapsto [1, 0], \qquad u \mapsto [0, 1]. \tag{1}$$

A length-n sequence is thus represented by a one-hot matrix $X \in \mathbb{R}^{n\times 2}$ with one token embedding per row. The learned query and key projections are $Q = XW_Q$ and $K = XW_K$ with $W_Q, W_K \in \mathbb{R}^{2\times d}$, so $Q, K \in \mathbb{R}^{n\times d}$ and q_{last} is the last row of Q. Here d is the query and key dimension, set to $d = 2$ throughout. Conventional self-attention forms the full $n \times n$ score matrix $QK^\top/\sqrt{d}$. This reduced classifier uses only the query at the last position and therefore needs only its scores against the n keys:

$$s_j = \frac{q_{\text{last}}^\top k_j}{\sqrt{d}}, \qquad j = 1, \ldots, n. \tag{2}$$

Since the token in the last position is u, $q_{\text{last}} = q_u$ for every example. Writing k_t and k_u for the target and non-target keys, which are position-independent, the target and non-target scores, and the score margin between them, are

$$a = \frac{q_u^\top k_t}{\sqrt{d}}, \qquad b = \frac{q_u^\top k_u}{\sqrt{d}}, \qquad \Delta = a - b. \tag{3}$$

The architecture thus fixes the score vector of a positive example to the two-score structure

$$S_n = (a, b, b, \ldots, b), \tag{4}$$

while training determines only the two values a and b. Because there is a single non-target type, every non-target position holds the same token, hence the same key and the same score b by construction.

The central change in this study is an additional score multiplier $\alpha(n)$ applied before softmax. The attention weight on position j is

$$A_j = \frac{e^{\alpha(n)s_j}}{\sum_{i=1}^{n} e^{\alpha(n)s_i}}. \tag{5}$$

Since $\alpha(n)$ multiplies the scores inside the softmax, it acts as a length-dependent inverse temperature: larger $\alpha(n)$ sharpens the attention. Standard attention corresponds to $\alpha(n) = 1$. The length-aware variants instead make $\alpha(n)$ grow with the sequence length. We consider three choices of $\alpha(n)$: constant, logarithmic, and learned logarithmic. Each is analyzed in turn below.

For the positive score vector $S_n = (a, b, \ldots, b)$, the attention mass on the single target key is

$$p_t(n) = \frac{e^{\alpha(n)a}}{e^{\alpha(n)a} + (n-1)e^{\alpha(n)b}} = \frac{e^{\alpha(n)\Delta}}{e^{\alpha(n)\Delta} + (n-1)}, \tag{6}$$

where the second form follows by dividing the numerator and denominator by $e^{\alpha(n)b}$.

The value pathway reuses these one-hot embeddings directly as value vectors, with no learned value projection, so the value vector v_j at position j is $[1, 0]$ when its token is the target and $[0, 1]$ otherwise. This makes the output directly record how much total attention is assigned to each token type. The classifier reads the weighted sum of these value vectors:

$$o(n) = \sum_{j=1}^{n} A_j v_j. \tag{7}$$

On a positive example, the target carries weight $p_t(n)$ and the non-targets together carry the remaining $1 - p_t(n)$, so the output is a convex combination of the two value vectors:

$$\begin{aligned} o_{\text{pos}}(n) &= p_t(n)[1,0] + \big(1 - p_t(n)\big)[0,1] \\ &= \big(p_t(n), 1 - p_t(n)\big). \end{aligned} \tag{8}$$

A negative example contains only u, so its output is always

$$o_{\text{neg}}(n) = [0,1]. \tag{9}$$

The negative output is this same combination with target mass 0, so both outputs are fixed by the single scalar $p_t(n)$. A learned linear layer reads this output, so on a positive example the classifier produces the logit

$$\begin{aligned} z(n) &= w_t p_t(n) + w_u\big(1 - p_t(n)\big) + \beta \\ &= (w_t - w_u)p_t(n) + (w_u + \beta), \end{aligned} \tag{10}$$

with learned weights w_t, w_u and bias β, and predicts target-present when $z(n) \geq 0$. Since the slope $w_t - w_u$ is positive in the trained model, the logit increases with $p_t(n)$, so classification reduces to a threshold: the model predicts target-present when $p_t(n) \geq p^*$, with $p^* = -(w_u + \beta)/(w_t - w_u)$. If $p_t(n) \to 0$, the positive representation converges to the negative representation, and the representation gap between the two classes vanishes. If $p_t(n)$ converges to a constant strictly between 0 and 1, classification depends on this threshold. The stronger, target-dominant outcome is $p_t(n) \to 1$. The conditions below distinguish these three cases.

3.1 Length-Scaling Regimes

Assume that $\Delta > 0$, so the target has a score advantage over each non-target. If $\Delta \leq 0$, none of the multipliers considered below can create such an advantage, since each $\alpha(n)$ is positive and scales Δ without changing its sign. The competing term $(n-1)$ is the source of length dependence: it grows like n, so the target term $e^{\alpha(n)\Delta}$ must grow at least as fast for target attention to survive. A logarithmic $\alpha(n)$ is the natural way to achieve this, since it turns $e^{\alpha(n)\Delta}$ into a power of n.

For the constant baseline $\alpha(n) = 1$,

$$p_t(n) = \frac{e^{\Delta}}{e^{\Delta} + (n-1)} \longrightarrow 0 \qquad \text{as } n \to \infty. \tag{11}$$

A fixed score margin can beat each non-target token individually, but it cannot beat an unbounded number of non-target competitors. Consequently, $o_{\text{pos}}(n) \to [0,1]$, the same representation as $o_{\text{neg}}(n)$.

For the log multiplier $\alpha(n) = \log n$, we have $e^{\alpha(n)\Delta} = e^{\Delta \log n} = n^{\Delta}$, so

$$p_t(n) = \frac{n^{\Delta}}{n^{\Delta} + n - 1}. \tag{12}$$

The limit depends on the score margin:

$$\lim_{n\to\infty} p_t(n) = \begin{cases} 1, & \Delta > 1, \\ \frac{1}{2}, & \Delta = 1, \\ 0, & 0 < \Delta < 1. \end{cases} \tag{13}$$

Thus $\Delta > 1$ makes target attention converge to 1, while $0 < \Delta < 1$ causes the positive representation to collapse onto the negative representation. At the boundary $\Delta = 1$, the two representations remain distinct, and classification depends on the learned decision boundary.

For the learned-log multiplier used in the experiments,

$$\alpha(n) = 1 + c\log(1+n), \tag{14}$$

where c is a learned, strictly positive coefficient. The additive 1 makes $\alpha = 1$ at $c = 0$, recovering the constant baseline, and using $\log(1+n)$ keeps the multiplier well-defined at small n; neither term changes the asymptotic exponent, which is governed by $c\Delta$. Since $e^{\alpha(n)\Delta} = e^{\Delta}(1+n)^{c\Delta}$, the target attention mass is

$$p_t(n) = \frac{e^{\Delta}(1+n)^{c\Delta}}{e^{\Delta}(1+n)^{c\Delta} + (n-1)}. \tag{15}$$

Its limit is

$$\lim_{n\to\infty} p_t(n) = \begin{cases} 1, & c\Delta > 1, \\ \dfrac{e^{\Delta}}{e^{\Delta}+1}, & c\Delta = 1, \\ 0, & c\Delta < 1. \end{cases} \tag{16}$$

Thus $c\Delta > 1$ makes target attention converge to 1. The equality case again has a nonzero limiting target mass and depends on the classifier's decision boundary, whereas $c\Delta < 1$ produces asymptotic collapse.

A single condition unifies all three regimes. Writing the target attention mass as

$$p_t(n) = \frac{1}{1+\exp(\log(n-1) - \alpha(n)\Delta)} = \sigma\big(g(n)\big), \qquad g(n) = \alpha(n)\Delta - \log(n-1), \tag{17}$$

with σ the sigmoid function, target attention converges to 1 exactly when $g(n) \to +\infty$, that is, when the effective margin $\alpha(n)\Delta$ outgrows $\log(n-1)$ without bound. Since $\log(n-1)$ grows like $\log n$ with coefficient 1, the outcome is decided by how fast $\alpha(n)\Delta$ grows: it stays bounded for the constant baseline, so target attention always collapses, and grows like $\Delta\log n$ for the log multiplier and $c\Delta\log n$ for the learned-log multiplier. Target dominance therefore needs the coefficient of $\log n$ to exceed 1, giving the thresholds $\Delta > 1$ and $c\Delta > 1$. This is the precise sense in which length generalization requires the effective margin to grow faster than the logarithm of the number of competing non-target keys.

4 Experimental Design

The experiments train the reduced model derived above (fixed one-hot embeddings reused as value vectors, learned query and key projections $W_Q, W_K \in \mathbb{R}^{2\times 2}$, a readout query, softmax attention, and a binary classifier on the two-dimensional attention output) and vary the score multiplier $\alpha(n)$ across three modes:

Mode	$\alpha(n)$
`constant`	1
`log`	$\log n$
`learned_log`	$1 + c\log(1+n)$

For `learned_log`, the coefficient is $c = \mathrm{softplus}(k_\alpha)$, where k_α is an unconstrained learnable scalar, so c stays positive during optimization.

The classifier maps the attention output to the logit $z(n)$ introduced above; an example is classified target-present when $z(n) \geq 0$, equivalently when its sigmoid probability is at least 0.5. Two quantities from the classifier appear in the results: the positive-example logit, the value of $z(n)$ on a positive example, whose sign decides whether that example is classified correctly; and the positive-example accuracy (equivalently, recall), the fraction of positive examples with $z(n) \geq 0$.

Table 1: Main results at $n = 10^7$, as means over five seeds ($\pm$ one standard deviation). The p_t, logit, and accuracy columns are measured on positive examples; negative accuracy is 100% in every run.

Run	Steps	Δ	c	$c\Delta$	p_t	Logit	Accuracy
`constant_e50`	1,600	9.0 ± 0.2	n/a	n/a	0.001 ± 0.000	-3.3 ± 0.1	0%
`constant_e100`	3,200	9.9 ± 0.2	n/a	n/a	0.002 ± 0.000	-4.5 ± 0.1	0%
`constant_e1000`	32,000	13.2 ± 0.2	n/a	n/a	0.050 ± 0.010	-17.3 ± 0.4	0%
`log_e50`	1,600	4.4 ± 0.1	n/a	n/a	1.000 ± 0.000	3.2 ± 0.1	100%
`learned_log_e50`	1,600	8.1 ± 0.2	0.072 ± 0.006	0.58 ± 0.03	0.788 ± 0.075	1.9 ± 0.5	100%
`learned_log_e100`	3,200	8.6 ± 0.3	0.096 ± 0.008	0.83 ± 0.05	0.997 ± 0.002	4.6 ± 0.1	100%
`learned_log_e200`	6,400	9.0 ± 0.3	0.126 ± 0.010	1.14 ± 0.06	1.000 ± 0.000	6.4 ± 0.1	100%
`learned_log_e400`	12,800	9.4 ± 0.3	0.166 ± 0.013	1.55 ± 0.08	1.000 ± 0.000	9.6 ± 0.1	100%

Each run trains at length 10 on 2,000 examples split evenly between positive and negative, minimizing binary cross-entropy on the logit with AdamW (learning rate 3×10^{-3}, no weight decay, batch size 64). One epoch is a single pass over the 2,000 examples, or 32 optimizer steps. Each configuration is trained under five random seeds (0–4), and all reported numbers are the mean and standard deviation across these seeds.

Each trained model is evaluated at the seven powers of ten from 10 to 10^7, six orders of magnitude beyond the training length, on 50 balanced examples per length. Because a length-10^7 evaluation cannot be held in memory at once, examples are generated and scored in small chunks, so the full evaluation tensor is never materialized.

Each run is labeled by its multiplier mode and epoch budget: for example, `constant_e50` is constant scaling trained for 50 epochs. Table 1 lists all eight runs: constant at 50, 100, and 1,000 epochs; log at 50; and learned-log at 50, 100, 200, and 400. Figure 1 shows six of them, keeping every constant budget and the two learned-log budgets that bracket the $c\Delta = 1$ threshold; the omitted `learned_log_e100` and `learned_log_e400` fall on the same sides.

5 Results

In this task, length generalization is decided entirely by the positive examples. A negative sequence contains only non-target values, so its attention output is exactly the non-target representation $[0, 1]$ by construction, independent of how attention is spread; because the learned threshold satisfies $p^* > 0$ in every run, negatives are classified correctly at every length. The difficulty lives in the positive examples: the target mass p_t can dilute with length until the classifier no longer detects the target. Table 1 reports each run at 10^7, and Figure 1 traces the target mass p_t in panel (a) and the positive-example logit in panel (b).

5.1 Constant Scaling

Constant scaling uses $\alpha = 1$. At any fixed Δ this gives $p_t(n) \to 0$. Empirically, more training increases Δ, moving the failure point outward without changing this asymptotic regime: the positive-example accuracy at 10^7 is 0 for all three budgets in every seed. More training even drives the positive-example logit more negative, not less (Table 1: -3.3, -4.5, and -17.3 for e50, e100, and e1000): once the target mass has collapsed, the logit is set by the learned intercept $w_u + \beta$, which itself grows more negative with training. The larger margin only postpones the collapse; the theory still predicts failure for any finite fixed margin.

5.2 Log Scaling

Log scaling uses $\alpha = \log n$. The score margin is $\Delta = 4.4 \pm 0.1$, comfortably above the threshold $\Delta > 1$ in every seed, so the theory predicts $p_t(n) \to 1$, which is what is observed: target attention reaches 1.000 at long lengths, the positive-example logit stays positive, and the positive-example accuracy is 1 at 10^7 in all five seeds.

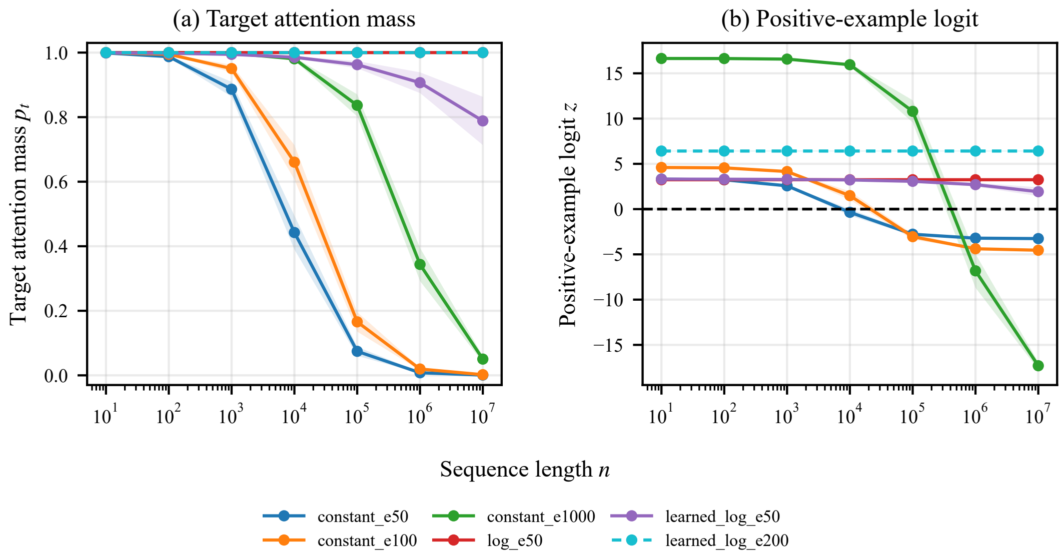

Figure 1: Target attention mass and positive-example logit versus sequence length for six representative runs (mean over five seeds; shaded bands show ± 1 s.d.). (a) Runs that generalize saturate at $p_t = 1$, so `log_e50` and `learned_log_e200` overlap, with the latter dashed. (b) The logit curves distinguish these overlapping runs; the horizontal dashed line at $z = 0$ marks the decision boundary.

5.3 Learned Log Scaling

Learned-log scaling uses $\alpha = 1 + c\log(1+n)$. These runs separate finite-length success from asymptotic success. At 50 and 100 epochs the model already passes the 10^7 benchmark, with positive-example accuracy 1 in all five seeds, yet the product $c\Delta$ stays below the threshold, 0.58 ± 0.03 and 0.83 ± 0.05, respectively, so the theory predicts eventual failure at larger lengths. Solving the closed-form $p_t(n)$ for the length at which target attention drops to the classifier's per-seed decision threshold p^* quantifies this (derivation in Appendix A): across seeds, the 50-epoch runs ($c\Delta \approx 0.58$) are predicted to fail near $n \sim 10^8$–10^9, within about two orders of magnitude of the benchmark. The 100-epoch runs ($c\Delta \approx 0.83$) are pushed beyond $n \sim 10^{18}$, the failure length climbing steeply as $c\Delta \to 1$.

Only at 200 epochs does $c\Delta$ exceed 1 in every seed (1.14 ± 0.06), entering the asymptotic-success regime; by 400 epochs it is well clear (1.55 ± 0.08). The crossing is driven by the coefficient c, which grows monotonically with the training budget ($0.072 \to 0.096 \to 0.126 \to 0.166$) while the score margin Δ grows only modestly ($8.1 \to 8.6 \to 9.0 \to 9.4$).

The e50 and e100 budgets reach 100% at 10^7 with $c\Delta < 1$: passing at one length does not certify generalization to every length. The property that transfers is $c\Delta > 1$, not accuracy at any single point. Table 1 shows this directly: `constant_e50` and `learned_log_e200` learn the same score margin ($\Delta = 9.0$), yet the first collapses ($p_t = 0.001$, 0%) and the second saturates ($p_t = 1.000$, 100%). The difference comes from the length scaling, not from Δ.

6 Mechanism: What the Model Learns

The reduced model also allows a direct weight-level explanation of where Δ comes from. Recall that the readout query is q_u (the last token is always the non-target u), the target key is k_t, and the non-target key is k_u. The target and non-target scores are then

$$a = \frac{q_u^\top k_t}{\sqrt{d}}, \qquad b = \frac{q_u^\top k_u}{\sqrt{d}}. \tag{18}$$

The score margin is therefore

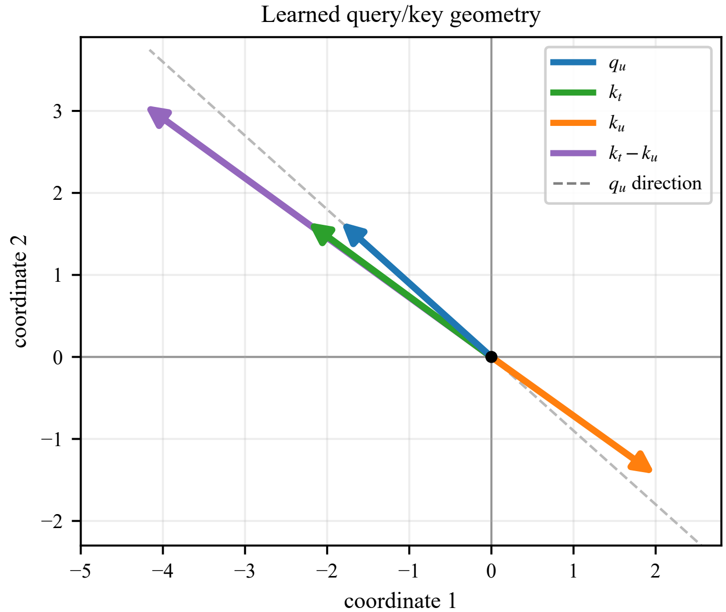

Figure 2: Learned query and key vectors for `learned_log_e200` (seed 1), drawn in the $d = 2$ query/key space.

$$\Delta = a - b = \frac{q_u^\top (k_t - k_u)}{\sqrt{d}}. \tag{19}$$

For one `learned_log_e200` checkpoint (seed 1), the learned vectors are approximately

$$q_u = \begin{bmatrix} -1.830 \\ 1.646 \end{bmatrix}, \qquad k_t = \begin{bmatrix} -2.232 \\ 1.642 \end{bmatrix}, \qquad k_u = \begin{bmatrix} 1.990 \\ -1.428 \end{bmatrix}. \tag{20}$$

With $d = 2$, this gives

$$a \approx 4.799, \qquad b \approx -4.237, \qquad \Delta \approx 9.036. \tag{21}$$

Geometrically (Figure 2), q_u points in a direction that separates the target key from the non-target key. The target key has a positive projection along the readout query direction, while the non-target key has a negative projection. Equivalently, q_u aligns with the difference vector $k_t - k_u$. This alignment creates the score pattern $a > b$, the positive score margin Δ that the length-scaling analysis assumes.

The geometry is the same in every seed: across seeds 0–4 the target score a is positive and the non-target score b negative, q_u is nearly collinear with $k_t - k_u$ (cosine ≥ 0.99), and $\Delta = 9.03 \pm 0.28$.

This mechanism explains how the model creates a target advantage, but it also shows why a target advantage is not by itself enough. Under constant scaling, even a large fixed Δ is eventually overwhelmed by the growing number of non-target positions. Learned-log attention has an additional degree of freedom: it can increase the coefficient c so that the effective margin grows like $c\Delta \log n$. In the successful learned-log run, optimization makes the product $c\Delta$ cross the threshold.

7 Discussion

The main contribution of this report is a controlled analysis of length-aware attention scaling in a trainable reduced model. The two-token, position-independent construction makes the two-score structure expected by design; it is not itself a learned discovery. This simplification lets us apply the closed-form target-attention equation directly and isolate the effect of the score multiplier from other transformer components.

The result also sharpens the distinction between finite and asymptotic generalization. A run can clear a fixed length such as 10^7 while $c\Delta$ still sits below 1, so passing a benchmark certifies only finite-length behavior, not the limit. What survives to unbounded length is a structural property of the learned solution, $c\Delta > 1$, and exposing that property cleanly, rather than a single-length benchmark score, is what the reduced model is for.

The reduced model is intentionally limited. It uses fixed semantic value vectors, no positional encodings, one readout query, and a simple binary classifier. These restrictions make the mechanism easy to analyze, but they also mean the conclusion should not be transferred directly to a full transformer, which can have non-identical non-target scores, learned value vectors, multiple heads, residual streams, feed-forward layers, layer normalization, and pooling. Any of these can break the exact two-score structure the analysis relies on.

The value of the reduced model is therefore diagnostic. It separates two questions that are entangled in a full transformer. First, can the architecture create a score margin Δ? Second, does the length-scaling mechanism make the effective margin grow fast enough to beat the softmax denominator? In this controlled binary setting, the answer to both questions can be measured directly.

Two directions follow. The first is to relax these restrictions toward a full transformer and test whether an analog of the $c\Delta > 1$ condition still governs length generalization once non-identical non-target scores, learned values, and depth are reintroduced. The second is to move beyond binary detection to broader versions of the task, such as identifying which target type is present or counting how many targets occur.

8 Conclusion

This report studied length generalization in a reduced binary attention classifier whose two-score structure makes the closed-form target-attention expression exact.

Across the three scaling rules, the experiments give a consistent picture: constant scaling fails at any sufficiently large length, while log and learned-log scaling generalize only when their effective margin outgrows the competing crowd, requiring $\Delta > 1$ and $c\Delta > 1$, respectively. In this reduced setting, then, the central condition for length generalization is not fitting the training length or passing a finite long-length benchmark; the effective margin must grow faster than the logarithm of the number of competing non-target keys.

A Supporting Material

A.1 Learned Query and Key Matrices for the Mechanism Example

The Mechanism section reports the query and key vectors of `learned_log_e200` (seed 1). They come from the two learned projection matrices, which for that run are

$$W_Q = \begin{pmatrix} 0.364 & -0.137 \\ -1.830 & 1.646 \end{pmatrix}, \qquad W_K = \begin{pmatrix} -2.232 & 1.642 \\ 1.990 & -1.428 \end{pmatrix}. \tag{22}$$

Because the token embeddings are one-hot, with $t \mapsto [1, 0]$ and $u \mapsto [0, 1]$, each token's query and key is the corresponding row: the first row gives q_t and k_t, and the second gives q_u and k_u. This recovers the vectors and scores reported in the Mechanism section.

Two details are worth recording. The readout query is always q_u, because the last token is always the non-target u, so the first row of W_Q is never used by the model; it stays near its initialization at $q_t = (0.364, -0.137)$. The matrices are also not individually identifiable: the model depends on them only through the product $W_Q W_K^\top$, whose entries divided by $\sqrt{d}$ are the scores, so any other pair with the same product defines exactly the same model. The invariant content is the scores, not the individual entries.

A.2 Predicted Learned-Log Failure Lengths

For a learned-log run with $c\Delta < 1$, the target attention decays to zero, so a positive example is misclassified once $p_t(n)$ falls to the classifier's decision threshold $p^* = -(w_u + \beta)/(w_t - w_u)$. We call the length at which this happens the failure length n^*, defined by $p_t(n^*) = p^*$. In every learned-log run the trained threshold sits close to one half ($p^* \approx 0.497$), so n^* is essentially the length at which the target loses the majority of the attention.

Substituting the learned-log closed form $p_t(n) = e^{\Delta}(1+n)^{c\Delta} / \big(e^{\Delta}(1+n)^{c\Delta} + (n-1)\big)$ and rearranging gives

$$\frac{n-1}{(1+n)^{c\Delta}} = e^{\Delta} \frac{1-p^*}{p^*}. \tag{23}$$

At the lengths of interest, $n - 1 \approx n$ and $1 + n \approx n$, so

$$n^* \approx \left(e^{\Delta} \frac{1-p^*}{p^*} \right)^{1/(1-c\Delta)}. \tag{24}$$

The exponent $1/(1 - c\Delta)$ is the important term. It diverges as $c\Delta \to 1$, so a small change in $c\Delta$ near the threshold moves n^* by many orders of magnitude.

Evaluating this per seed, using each run's own Δ, c, and p^*, gives failure lengths of $10^{7.7}$ to $10^{9.2}$ for `learned_log_e50` ($c\Delta \approx 0.58$) and $10^{17.8}$ to $10^{34.4}$ for `learned_log_e100` ($c\Delta \approx 0.83$). Both budgets clear the 10^7 benchmark, but the 50-epoch runs are predicted to fail within about two orders of magnitude of it, while the 100-epoch runs fail only far beyond it. The approximation above agrees with a direct numerical solve of $p_t(n^*) = p^*$ to the precision shown.

Use of Artificial Intelligence

AI assistants were used throughout this project. They wrote most of the experiment code, including the model and the training loop. They organized the run data into a readable form and assisted the analysis that produced the diagnostics behind Table 1, the failure-length prediction in Appendix A, and the figures. They also drafted and revised the report prose.

The direction of the work was human throughout. The author ran the experiments, derived the closed-form analysis and the $c\Delta > 1$ condition, chose the questions, directed the AI's work at every step, and made the judgment calls that shaped the report. The reduced model was proposed by the project advisor, who also reviewed the manuscript and set the scope of the main text. Every reported number was checked against the run data, and the author is responsible for all claims made here.

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